CSE STUDY HUB

Communication Skills for Engineers II — Aptitude

UNIT 3 · Percentage Applications

What this unit is about

This unit builds understanding of **Percentage Applications**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Percentage as a fraction out of 100 and conversion to ratio

2. Finding a percentage of a quantity and reverse percentage

3. Base value and change of base

4. Percentage increase/decrease and multiplying factor

5. Successive percentage changes: multiply factors, do not simply add

6. Population, salary, marks, price and expenditure word problems

7. Comparing percentage change with absolute change

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 3 NOTES · PART 1

Percentage as a fraction out of 100 and conversion to ratio

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Finding a percentage of a quantity and reverse percentage

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Base value and change of base

Base value and change of base is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Percentage Applications, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 2

Percentage increase/decrease and multiplying factor

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Successive percentage changes: multiply factors, do not simply add

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Percentage Applications, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 3

Population, salary, marks, price and expenditure word problems

Population, salary, marks, price and expenditure word problems is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Comparing percentage change with absolute change

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Percentage Applications, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Percentage as a fraction out of 100 and conversion to ratio** — know the definition, core working and one example.

• **Finding a percentage of a quantity and reverse percentage** — know the definition, core working and one example.

• **Base value and change of base** — know the definition, core working and one example.

• **Percentage increase/decrease and multiplying factor** — know the definition, core working and one example.

• **Successive percentage changes: multiply factors, do not simply add** — know the definition, core working and one example.

• **Population, salary, marks, price and expenditure word problems** — know the definition, core working and one example.

• **Comparing percentage change with absolute change** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Percentage Applications. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Percentage as a fraction out of 100 and conversion to ratio and explain its importance.

Q2. Explain the working/steps involved in Finding a percentage of a quantity and reverse percentage.

Q3. Differentiate between two related concepts from Percentage Applications.

Q4. Give one practical application of Comparing percentage change with absolute change and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Percentage Applications.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.